



Linear Regression 101: Bevington Archetype

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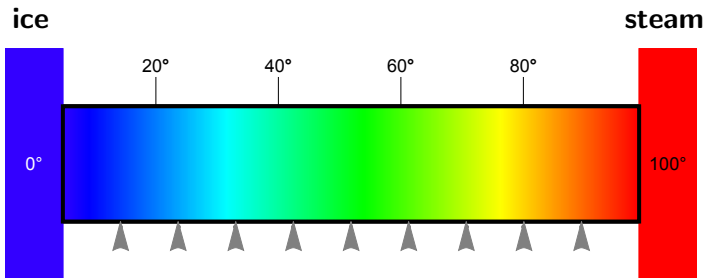
Defense Threat Reduction Information Analysis Center (DTRIAC)

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Linear Regression: Heat Flow



Compute the temperature of the bar as a function of position.



Overview

- 1 Problem
- 2 Model
- 3 Solution
- 4 Result
- 5 Discussion
- 6 Tinker

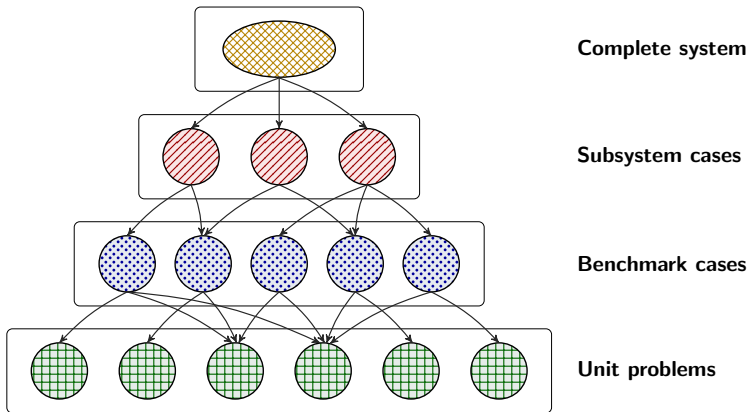


Demonstration Problem

Let's pose and solve a classic **least squares** problem



Validation Tiers of a System Hierarchy





Problem
Model
Solution
Result
Discussion
Tinker

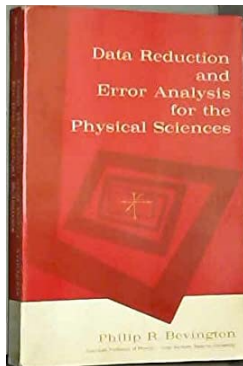
Bevington Archetype
Least Squares Exemplar Problem
Fundamental Theorem of Linear Algebra
Least Squares Problem and Solution
The Fredholm Alternative



Canonical Reference: Bevington, 2e

Data Reduction and Error Analysis for the Physical Sciences, 2e Philip Bevington

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Bevington Table 6.1

Trial i	Position x_i , cm	Temperature T_i , °C	x_i^2	$x_i T_i$	T_i^2	$a + bx_i$	Δ_i^2
1	1.0	15.6	1.0	15.6	243.36	14.2	1.96
2	2.0	17.5	4.0	35.0	306.25	23.6	37.21
3	3.0	36.6	9.0	109.8	1339.56	33.0	12.96
4	4.0	43.8	16.0	175.2	1918.44	42.4	1.96
5	5.0	58.2	25.0	291.0	3387.24	51.8	40.96
6	6.0	61.6	36.0	369.6	3794.56	61.2	0.16
7	7.0	64.2	49.0	449.4	4121.64	70.6	40.96
8	8.0	70.4	64.0	563.2	4956.16	80.0	92.16
9	9.0	98.8	81.0	889.2	9761.44	89.4	88.36
SUM	45.0	466.7	285.0	2898.0	29,828.65		316.69

$$\Delta = N \Sigma x_i^2 - (\Sigma x_i)^2 = 9(285) - (45)^2 = 540$$

$$a = \frac{\Sigma x_i^2 \Sigma T_i - \Sigma x_i \Sigma x_i T_i}{\Delta} = \frac{285(466.7) - 45(2898)}{540} = 4.8$$

$$b = \frac{N \Sigma x_i T_i - \Sigma x_i \Sigma T_i}{\Delta} = \frac{9(2898) - 45(466.7)}{540} = 9.4$$

$$s^2 = \frac{1}{N-2} \Sigma (T_i - a - bx_i)^2 = \frac{1}{N-2} \Sigma (\Delta_i)^2 = \frac{1}{7}(316.69) = 45.24$$

$$s^2 = \frac{1}{N-2} [\Sigma T_i^2 + Na^2 + b^2 \Sigma x_i^2 - 2(a \Sigma T_i - ab \Sigma x_i + b \Sigma x_i T_i)]$$

$$= \frac{1}{7} [29,828.65 + 207.36 + 25,182.60 - 2(2240.16 - 2030.40 + 27,241.2)] = 45.24$$

$$\sigma_a \approx \frac{s^2 \Sigma x_i^2}{\Delta} = 23.9 \quad \sigma_a \approx 4.9^\circ\text{C}$$

$$\sigma_b \approx \frac{Ns^2}{\Delta} = 0.754 \quad \sigma_b \approx 0.87^\circ\text{C/cm}$$

$$\sigma \approx \sqrt{s^2} = 6.7 \quad \text{P.E.} = \frac{2}{\sqrt{5}} \sigma \approx 4.5^\circ\text{C}$$



Bevington Table 6.1

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$$s^2 = \frac{1}{N-2} [\sum T_i^2 + N a^2 + b^2 \sum x_i^2 - 2(a \sum T_i + b \sum x_i T_i)]$$

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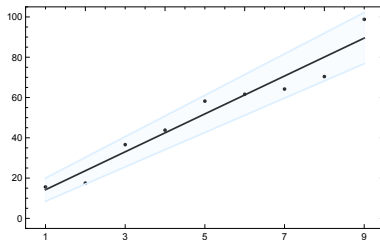
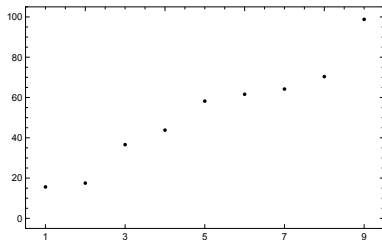


What we have and what we want

input



output



$\{\text{position, temperature}\}_{k=1}^m$



intercept and slope

$\{x_k, T_k\}_{k=1}^m$



$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix}$$



The Method of Least Squares.

The method of least squares

- **extends toolkit** of linear algebra to inconsistent systems
- provides **quantification of the error** in the fit parameters



Least Squares Exemplar Problem

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \quad (1)$$



Least Squares Exemplar Problem

Problem:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \quad (1)$$



Least Squares Exemplar Solution

Problem:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \textcolor{red}{0} & \textcolor{red}{0} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \textcolor{red}{y_3} \end{bmatrix} \quad (1)$$

Solution:

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad (2)$$



Least Squares Exemplar Error

Problem:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ 0 \end{bmatrix} \quad (1)$$

Solution:

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad (2)$$

Residual error:

$$r^2 = \left\| \begin{bmatrix} y_1 \\ y_2 \\ 0 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \right\|_2^2 = |y_3|^2 \quad (3)$$



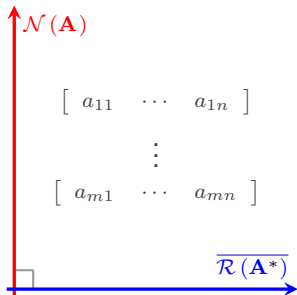
Fundamental Theorem of Linear Algebra

The Fundamental Theorem of Linear Algebra

$$\mathbf{A} \in \mathbb{C}^{m \times n}$$

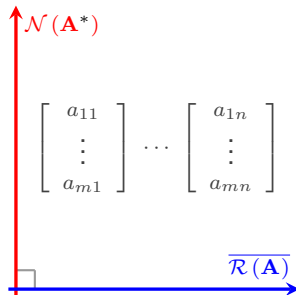
$$\mathbf{A}: \mathbb{C}^n \mapsto \overline{\mathcal{R}(\mathbf{A})}$$

$$\overline{\mathcal{R}(\mathbf{A}^*)} \hookleftarrow \mathbb{C}^m: \mathbf{A}^*$$



Solution Space

$$\mathbb{C}^n = \overline{\mathcal{R}(\mathbf{A}^*)} \oplus \mathcal{N}(\mathbf{A})$$



Data Space

$$\mathbb{C}^m = \overline{\mathcal{R}(\mathbf{A})} \oplus \mathcal{N}(\mathbf{A}^*)$$



Least Squares Untangles The Null Spaces



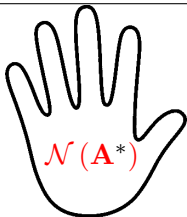
Figure: Vectors in **range spaces** ($\mathcal{R}(\mathbf{A})$ or $\mathcal{R}(\mathbf{A}^*)$) are colored in blue. Vectors in **null spaces** ($\mathcal{N}(\mathbf{A}^*)$ or $\mathcal{N}(\mathbf{A})$) are colored in red. Amalgam vectors with components in both spaces are the traditional black.



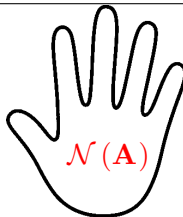
Null Space Nomenclature.

Left-hand null space

Right-hand null space



$\mathcal{R}(\mathbf{A})^\perp$
 $\text{null}(\mathbf{A}^*)$
 $\text{ker}(\mathbf{A}^*)$
cokernel



$\mathcal{R}(\mathbf{A}^*)^\perp$
 $\text{null}(\mathbf{A})$
 $\text{ker}(\mathbf{A})$
kernel



Problem Statement

Consider the linear equation

$$\mathbf{A} x = b$$

- 1 $\mathbf{A} \in \mathbb{C}_{\rho}^{m \times n}$: design matrix
- 2 $b \in \mathbb{C}^m$: data vector
- 3 $x \in \mathbb{C}^n$: solution vector



Least Squares Solution

The **Least Squares Solution** is defined as

$$x_{LS} = \arg \min_{x \in \mathbb{C}^n} \|\mathbf{A} x - b\|_2$$



Least Squares Solution

The **Least Squares Solution** is expressed as

$$\begin{aligned}
 x_{LS} &= \mathbf{A}^\dagger b & + & (\mathbf{I}_n - \mathbf{A}^\dagger \mathbf{A})y \\
 &= \mathbf{A}^\dagger b & + & \mathbf{P}_{\mathcal{N}(\mathbf{A})}y \\
 &= x_p & + & x_h \\
 &= x_{\mathcal{R}(\mathbf{A}^*)} & + & x_{\mathcal{N}(\mathbf{A})}
 \end{aligned} \tag{4}$$



Dramatis Personae

Least Squares Residual Error Vector

$$r(x) = \mathbf{A}x - b \quad (5)$$

Least Squares Merit Function

$$M(x) = \langle r(x), r(x) \rangle = \langle \mathbf{A}x - b, \mathbf{A}x - b \rangle \quad (6)$$

Least Total Squared Error

$$t^2 = \langle r(x_{LS}), r(x_{LS}) \rangle \quad (7)$$

Least Squares Uncertainty (Standard Deviation)

$$\sigma_k^2 = \frac{t^2}{m - n} (\mathbf{A}^* \mathbf{A})_{k,k}^{-1}, \quad k = 1:n \quad (8)$$



A Vital Theorem

Theorem (Fredholm Alternative)

Given a matrix $A \in \mathbb{C}^{m \times n}$, and a data vector $b \in \mathbb{C}^m$, either

- I. $Ax = b$ has a solution x , or
- II. $A^*y = 0$ has a solution y such that $y^*b \neq 0$.



A Vital Theorem

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Given a matrix $A \in \mathbb{C}^{m \times n}$, and a data vector $b \in \mathbb{C}^m$, either

I. $Ax = b$ has a solution x , or

II. $A^*y = 0$ has a solution y such that $y^*b \neq 0$.

Colloquially, **either** $Ax - b = 0$ **or** $Ax - b > 0$



Fredholm Alternative: Colloquially

Fredholm's Alternatives are

either $\mathbf{A} x - b = 0$

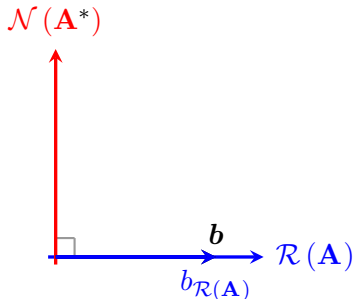
or $\mathbf{A} x - b > 0$



Fredholm Alternative in Pictures

either

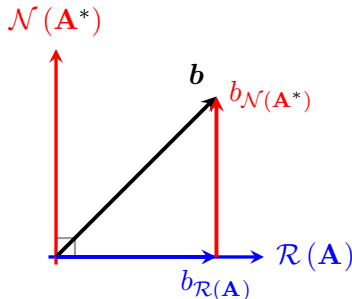
$$b = b_{\mathcal{R}(\mathbf{A})}$$



$\mathbf{A}x = b$ has solution

or

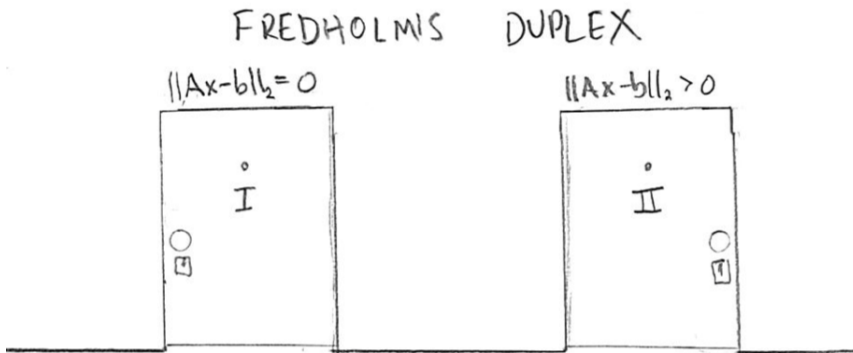
$$b = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}$$



$\mathbf{A}^*y = 0$ and $y^*b \neq 0$

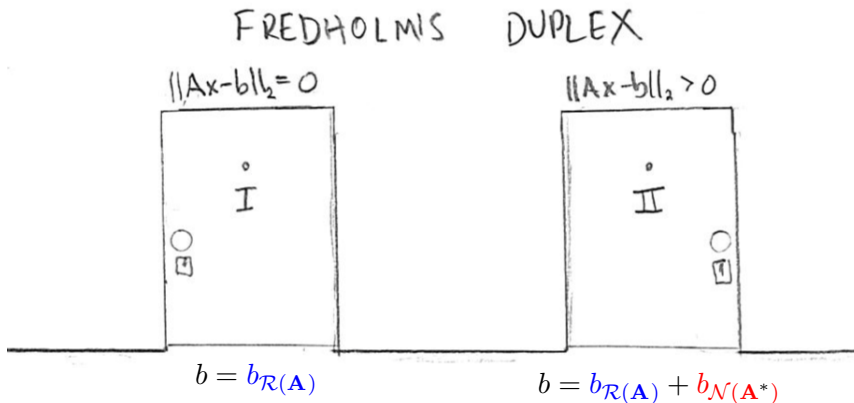


Fredholm's Duplex: Residency of the Data Vector





Fredholm's Duplex: Residency of the Data Vector





The Model

The **model** is an **input** to the least squares problem
along with the data.



The Model

Let's **derive the model** from first principles.



Heat Conduction Equation

Partial Differential Equation

$$T_t(x, t) = \alpha \Delta T(x, t) \quad (9)$$

Dirichlet Boundary Conditions

$$\begin{aligned} T(x = 0, t) &= 0, \\ T(x = 100, t) &= 100, \end{aligned} \quad (10)$$



Stationary Solution for Heat Conduction

$$\lim_{t \rightarrow \infty} T_t(x, t) = 0 \quad (11)$$



Laplace's Equation

The stationary solution for the **Heat Conduction**...

$$T_t(x, t) = \alpha \Delta T(x, t) \quad (12)$$

$$T_t = 0,$$



Laplace's Equation

The stationary solution for the **Heat Conduction**...

$$T_t(x, t) = \alpha \Delta T(x, t) \quad (12)$$

$$T_t = 0,$$

...is given by **Laplace's Equation**

$$0 = \alpha \Delta T(x, t) \quad (13)$$



Solve Using Integration

$$0 = \alpha \Delta T(x, t), \quad 0 \leq x \leq 10 \quad (14)$$

$$T(x) = 0 + 100x \quad (15)$$



Linear Model with Free Variables

Let's generalize the solution

$$\begin{aligned} T(x) &= 0 + 100x \\ &\Downarrow \\ T(x) &= a_0 + a_1x \end{aligned} \tag{16}$$



Model of Heat Flow

Temperature as a **linear function** of position

$$T(x) = a_0 + a_1x \quad (17)$$

Solution parameters:

a_0 intercept

a_1 slope

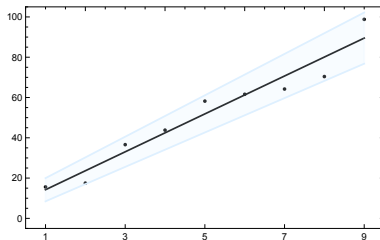
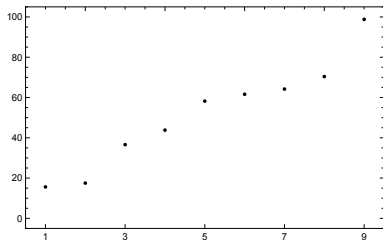


Least Squares Inputs and Outputs

input



output



$\{\text{position, temperature}\}_{k=1}^m$



intercept and slope

$\{x_k, T_k\}_{k=1}^m$



$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix}$$



Least Squares Inputs and Outputs

Inputs:

- 1 Model function, $T(x) = a_0 + a_1x$
- 2 Sequence of measurements, $\{x_k, T_k\}_{k=1}^m$



Least Squares Inputs and Outputs

Inputs:

- 1 Model function, $T(x) = a_0 + a_1x$
- 2 Sequence of measurements, $\{x_k, T_k\}_{k=1}^m$

Outputs: Model parameters

- 1 $a_0 \pm \sigma_0$, intercept with error
- 2 $a_1 \pm \sigma_1$, slope with error



Linear System

$$\mathbf{A} \quad a \quad = \quad T$$
$$\begin{bmatrix} \mathbf{1} & x \end{bmatrix} \quad \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \quad = \quad \begin{bmatrix} T \end{bmatrix}$$



Linear System

$$\begin{aligned} & \mathbf{A} \quad \mathbf{a} = \mathbf{T} \\ & \begin{bmatrix} 1 & x \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} T \end{bmatrix} \\ & \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 15.6 \\ 17.5 \\ 36.6 \\ 43.8 \\ 58.2 \\ 61.6 \\ 64.2 \\ 70.4 \\ 98.8 \end{bmatrix} \end{aligned} \tag{18}$$



Linear System: Dramatis Personae

$$\begin{matrix} & \mathbf{A} & & \mathbf{a} & = & T \\ \left[\begin{array}{cc} \mathbf{1} & x \end{array} \right] & & \left[\begin{array}{c} a_0 \\ a_1 \end{array} \right] & = & \left[\begin{array}{c} T \end{array} \right] \end{matrix} \quad (19)$$

- ① $\mathbf{A} \in \mathbb{R}_2^{9 \times 2}$: design matrix– mesh
- ② $b \in \mathbb{R}^9$: data vector
- ③ $a \in \mathbb{R}^2$: solution vector



The Matrix as a Map

The matrix A is a map from the
solution space $\mathbb{R}^2$ to the measurement space $\mathbb{R}^9$.



The Matrix as a Map

The matrix A is a map from the
solution space $\mathbb{R}^2$ to the measurement space $\mathbb{R}^9$.

We need the **inverse map**.



Notation: Inner Product

Given $x, y \in \mathbb{C}^m$

$$\begin{aligned}\langle x, y \rangle &= x^* y = x \cdot y = \sum \bar{x}_k y_k \\ &= \int_{\Omega} \bar{x} y d\Omega\end{aligned}\tag{20}$$



Normal Equations.

$$\begin{aligned} \mathbf{A}^* \mathbf{A} \quad a &= \mathbf{A}^* T \\ \begin{bmatrix} \langle \mathbf{1}, \mathbf{1} \rangle & \langle \mathbf{1}, x \rangle \\ \langle x, \mathbf{1} \rangle & \langle x, x \rangle \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} &= \begin{bmatrix} \langle \mathbf{1}, T \rangle \\ \langle x, T \rangle \end{bmatrix} \\ \begin{bmatrix} 9 & 45 \\ 45 & 285 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} &= \begin{bmatrix} 466.7 \\ 2898 \end{bmatrix} \end{aligned} \quad (21)$$



What Are We Minimizing?

Residual error vector:

$$r(a) = \mathbf{A}a - T \quad (22)$$



What Are We Minimizing?

Residual error vector:

$$r(a) = \mathbf{A}a - T \quad (22)$$

Minimize length of **residual error** vector:

$$\begin{aligned} M(a) &= \langle r(a), r(a) \rangle \\ &= \|\mathbf{A}a - T\|_2^2 \\ &= \|a_0 \mathbf{1} + a_1 x - T\|_2^2 \end{aligned} \quad (23)$$



What Are We Minimizing?

Residual error vector:

$$r(a) = \mathbf{A}a - T \quad (22)$$

Minimize length of **residual error** vector:

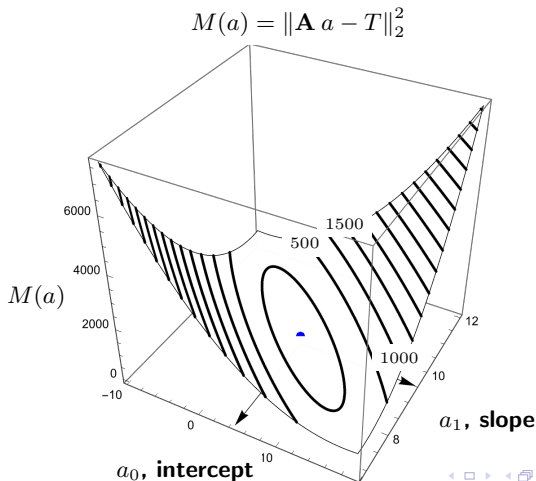
$$\begin{aligned} M(a) &= \langle r(a), r(a) \rangle \\ &= \|\mathbf{A}a - T\|_2^2 \\ &= \|a_0 \mathbf{1} + a_1 x - T\|_2^2 \end{aligned} \quad (23)$$

Archaic (scalar) notation:

$$M(a) = \sum_{k=1}^m (a_0 + a_1 x_k - T_k)^2$$



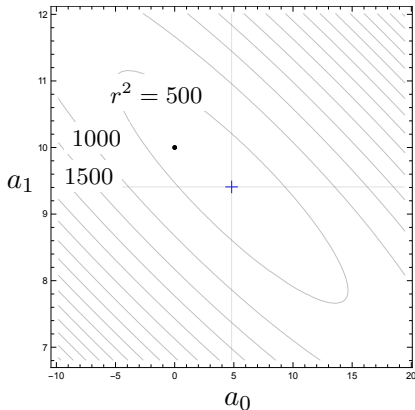
Minimization Target: Merit Function





Solution Space $\mathcal{R}(A^*)$: Bullseye

$$M(a) = \|a_0 \mathbf{1} + a_1 x - T\|_2^2$$





hierophant

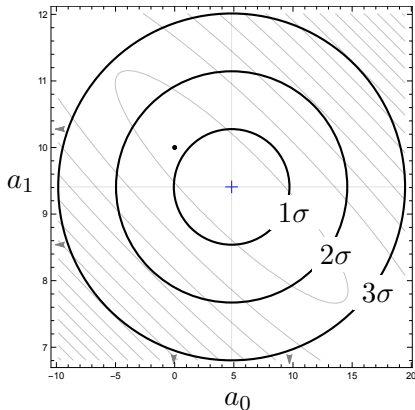
hi·er·o·phant | 'hī(ə)rəˌfant

a person, especially a priest in ancient Greece, who interprets sacred mysteries or esoteric principles.



Solution Space $\mathcal{R}(A^*)$: Bullseye

$$M(a) = \|a_0 \mathbf{1} + a_1 x - T\|_2^2$$





Numeric Solution

$$\begin{aligned}\sigma_0 &= 4.886206312183354 && \text{rounded to 4.9,} \\ a_0 &= 4.813888888888889 && \text{rounded to 4.8;} \\ \sigma_1 &= 0.8683016476563611 && \text{rounded to 0.87,} \\ a_1 &= 9.408333333333333 && \text{rounded to 9.41.}\end{aligned}\tag{24}$$



Numeric Solution

Numeric solution:

$$a_p = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix}$$
$$\sigma = \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix}_p = \begin{bmatrix} 4.9 \\ 0.87 \end{bmatrix}$$



Numeric Solution: Options

Either

$$a_p = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix} = \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix} \pm \begin{bmatrix} 4.9 \\ 0.87 \end{bmatrix} \quad (25)$$



Numeric Solution: Options

Either

$$a_p = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix} = \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix} \pm \begin{bmatrix} 4.9 \\ 0.87 \end{bmatrix} \quad (25)$$

Or

$$a_p = \begin{bmatrix} a_0 \pm \sigma_0 \\ a_1 \pm \sigma_1 \end{bmatrix}_p = \begin{bmatrix} 4.8 \pm 4.9 \\ 9.41 \pm 0.87 \end{bmatrix} \quad (26)$$



Bevington Table 6.1

Trial	Position	Temperature					
i	x_i , cm	T_i , °C	x_i^2	$x_i T_i$	T_i^2	$a + bx_i$	Δ_i^2
1	1.0	15.6	1.0	15.6	243.36	14.2	1.96
2	2.0	17.5	4.0	35.0	306.25	23.6	37.21
3	3.0	36.6	9.0	109.8	1339.56	33.0	12.96
4	4.0	43.8	16.0	175.2	1918.44	42.4	1.96
5	5.0	58.2	25.0	291.0	3387.24	51.8	40.96
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7	7.0	64.2	49.0	449.4	4121.64	70.6	40.96
8	8.0	70.4	64.0	563.2	4956.16	80.0	92.16
9	9.0	98.8	81.0	889.2	9761.44	89.4	88.36
SUM	45.0	466.7	285.0	2898.0	29,828.65		316.69

$$\Delta = N \Sigma x_i^2 - (\Sigma x_i)^2 = 9(285) - (45)^2 = 540$$

$$a = \frac{\Sigma x_i^2 \Sigma T_i - \Sigma x_i \Sigma x_i T_i}{\Delta} = \frac{285(466.7) - 45(2898)}{540} = 4.8$$

$$b = \frac{N \Sigma x_i T_i - \Sigma x_i \Sigma T_i}{\Delta} = \frac{9(2898) - 45(466.7)}{540} = 9.4$$

$$s^2 = \frac{1}{N-2} \Sigma (T_i - a - bx_i)^2 = \frac{1}{N-2} \Sigma (\Delta_i)^2 = \frac{1}{7}(316.69) = 45.24$$

$$s^2 = \frac{1}{N-2} [\Sigma T_i^2 + Na^2 + b^2 \Sigma x_i^2 - 2(a \Sigma T_i - ab \Sigma x_i + b \Sigma x_i T_i)]$$

$$= \frac{1}{7} [29,828.65 + 207.36 + 25,182.60 - 2(2240.16 - 2030.40 + 27,241.2)] = 45.24$$

$$\sigma_a \approx \frac{s^2 \Sigma x_i^2}{\Delta} = 23.9 \quad \sigma_a \approx 4.9^\circ\text{C}$$

$$\sigma_b \approx \frac{Ns^2}{\Delta} = 0.754 \quad \sigma_b \approx 0.87^\circ\text{C/cm}$$

$$\sigma \approx \sqrt{s^2} = 6.7 \quad \text{P.E.} = \frac{2}{\sqrt{5}} \sigma \approx 4.5^\circ\text{C}$$



Exact Solution

$$a_p = \Delta^{-1} \begin{bmatrix} \langle \mathbf{1}, T \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, T \rangle \\ \langle \mathbf{1}, \mathbf{1} \rangle \langle x, T \rangle - \langle x, \mathbf{1} \rangle \langle \mathbf{1}, T \rangle \end{bmatrix} \quad (27)$$

$$\Delta = \langle \mathbf{1}, \mathbf{1} \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, \mathbf{1} \rangle \quad (28)$$

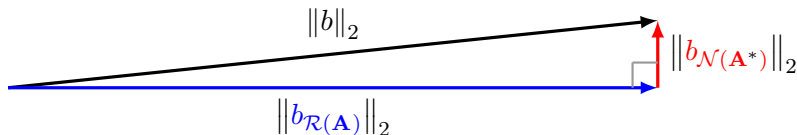


Exact Solution.

$$\begin{aligned}a_p &= \Delta^{-1} \begin{bmatrix} \langle \mathbf{1}, T \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, T \rangle \\ \langle \mathbf{1}, \mathbf{1} \rangle \langle x, T \rangle - \langle x, \mathbf{1} \rangle \langle \mathbf{1}, T \rangle \end{bmatrix} \\&= \frac{1}{360} \begin{bmatrix} 1733 \\ 3387 \end{bmatrix} \approx \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix}\end{aligned}$$

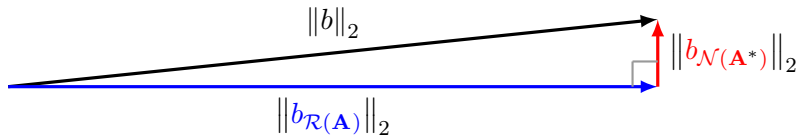


Resolution of Data Vector





Signal to Noise



$$\mathbb{S} = \frac{\|b_{\mathcal{R}(A)}\|_2}{\|b_{\mathcal{N}(A^*)}\|_2} = \frac{171.791}{17.795} \approx 9.65. \quad (29)$$



Posed and Solved

Problem posed: $\mathbf{A}a = T$

Problem solved: $\mathbf{A}a_p = T_{\mathcal{R}(\mathbf{A})}$

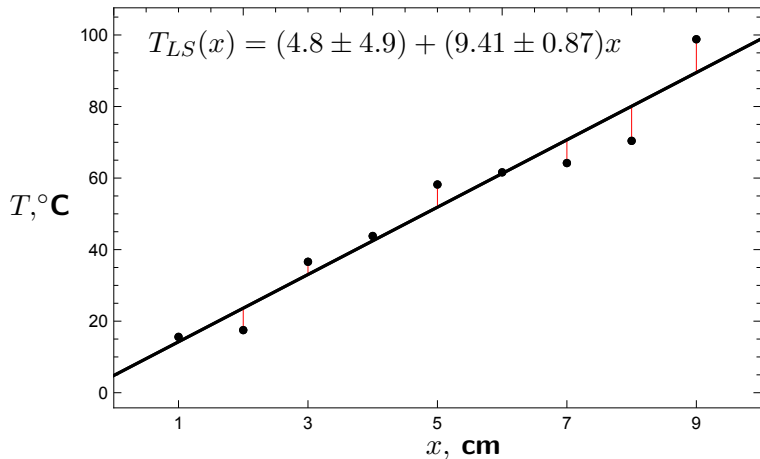


Resolving The Data Vector

$$\begin{aligned}
 \mathbf{A} \quad a_p &= T_{\mathcal{R}(\mathbf{A})} = T - T_{\mathcal{N}(\mathbf{A})} \\
 \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix} &= \frac{1}{360} \begin{bmatrix} 5120 \\ 8507 \\ 11894 \\ 15281 \\ 18668 \\ 22055 \\ 25442 \\ 28829 \\ 32216 \end{bmatrix} = \frac{1}{360} \begin{bmatrix} 5616 \\ 6300 \\ 13176 \\ 15768 \\ 20952 \\ 22176 \\ 23112 \\ 25344 \\ 35568 \end{bmatrix} - \frac{1}{360} \begin{bmatrix} 496 \\ -2207 \\ 1282 \\ 487 \\ 2284 \\ 121 \\ -2330 \\ -3485 \\ 3352 \end{bmatrix} \\
 &= 171.791^2 = 172.710^2 - 17.795^2 \quad (30)
 \end{aligned}$$

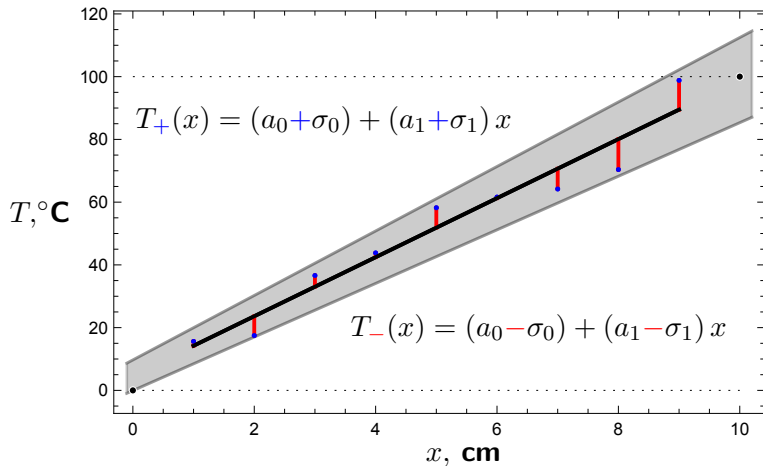


Data, Fit, Result, Residuals



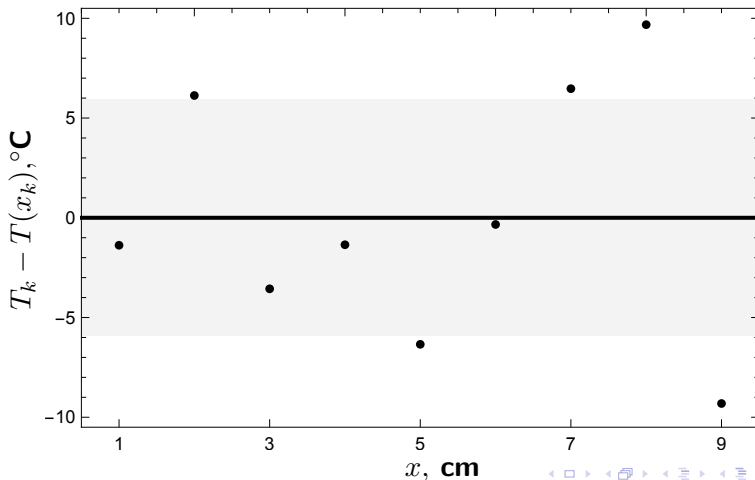


Data, Fit, 1 σ Error





Residual Errors: Correlation?





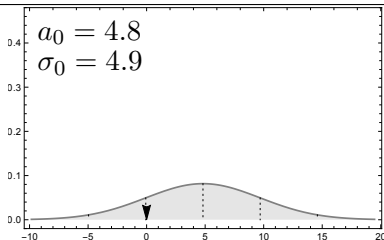
Normal Distribution

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (31)$$



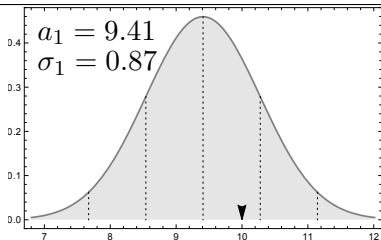
Solution parameters as normal distributions

intercept



less accurate
less precise

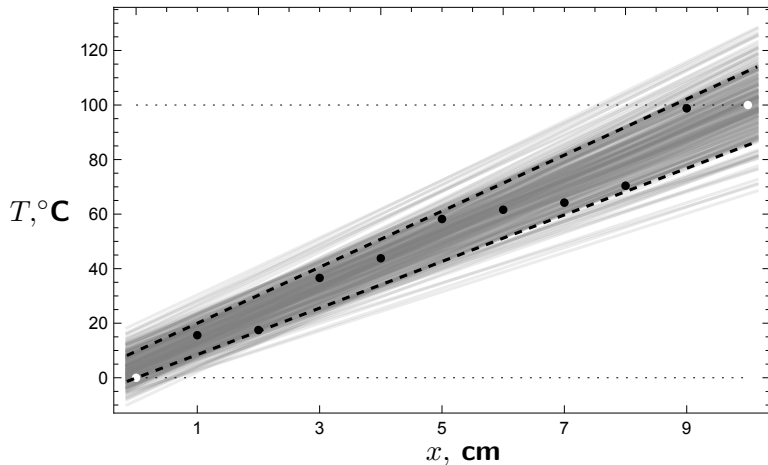
slope



more accurate
more precise



Data, Whiskers: 250 Random Samples.





Prediction Error

We know the temperature of the boiling water. What is the **uncertainty in the extrapolation** to $x = 10$ cm?

$$\sigma_T^2(x) = \sigma_0^2 + x^2 \sigma_1^2 + a_1^2 \sigma_x^2 \quad (32)$$



Prediction With Error

$$\begin{aligned} T(x) \pm \sigma_T &= (a_0 \pm \sigma_0) + (a_1 \pm \sigma_1) (x \pm \sigma_x) \\ T(10) \pm \sigma_T &= (4.8 \pm 4.9) + (9.41 \pm 0.87) (10 \pm \sigma_x) \end{aligned} \quad (33)$$



Prediction With Error

$$\begin{aligned}T(x) \pm \sigma_T &= (a_0 \pm \sigma_0) + (a_1 \pm \sigma_1) (x \pm \sigma_x) \\T(10) \pm \sigma_T &= (4.8 \pm 4.9) + (9.41 \pm 0.87) (10 \pm \sigma_x)\end{aligned}\tag{33}$$

$$\begin{aligned}T(x) &= a_0 + a_1 x, \\T(10) &= 4.8 + 9.41 \cdot 10 = 98.9.\end{aligned}\tag{34}$$



Prediction With Error

$$\begin{aligned} T(x) \pm \sigma_T &= (a_0 \pm \sigma_0) + (a_1 \pm \sigma_1) (x \pm \sigma_x) \\ T(10) \pm \sigma_T &= (4.8 \pm 4.9) + (9.41 \pm 0.87) (10 \pm \sigma_x) \end{aligned} \quad (33)$$

$$\begin{aligned} T(x) &= a_0 + a_1 x, \\ T(10) &= 4.8 + 9.41 \cdot 10 = 98.9. \end{aligned} \quad (34)$$

$$\begin{aligned} \sigma_T^2 &= \sigma_0^2 + x^2 \sigma_1^2 + a_1^2 \sigma_x^2, \\ \sigma_T^2 &= 4.886^2 + 10^2 \cdot 0.8683^2 + 9.41^2 \cdot 0^2 = 99.268. \end{aligned} \quad (35)$$



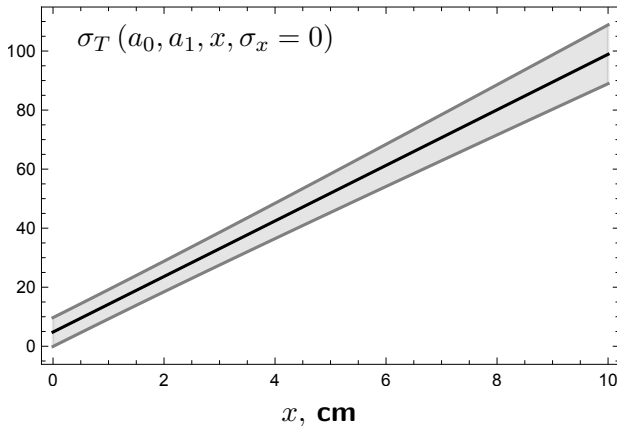
Final Prediction

$$\sigma_T = \sqrt{99.268} = 9.9631. \quad (36)$$

$$T(10) = (99. \pm 10.)^\circ C \quad (37)$$



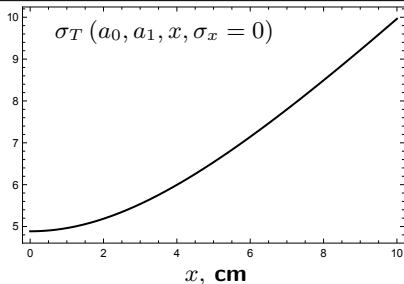
Prediction Errors



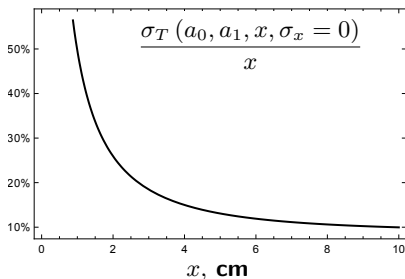


Making Predictions

Interpolation error, absolute



Interpolation error, relative





Refine The Mesh

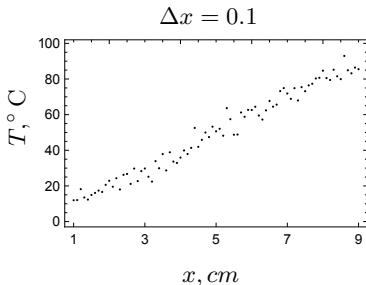
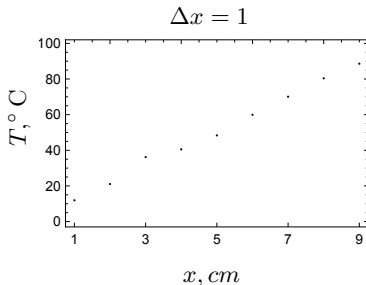
mesh size	measurements	a_0	a_1
1.	9	3.1 ± 1.5	9.53 ± 0.27
0.1	81	-0.4 ± 1.0	10.03 ± 0.18
0.01	801	-0.12 ± 0.34	10.011 ± 0.061
0.001	8 001	0.01 ± 0.11	9.990 ± 0.019
0.000 1	80 001	-0.029 ± 0.034	$10.004 \pm 0.006 1$
0.000 01	800 001	-0.001 ± 0.011	$9.999 4 \pm 0.001 9$
0.000 001	8 000 001	-0.0034 ± 0.0034	$10.000 54 \pm 0.000 61$
0.000 000 1	80 000 001	$0.001 160 8 \pm 0.000 002 9$	$9.999 891 \pm 0.000 094$



Mesh Refinement: Steps 1–2

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} 3.1 \\ 9.53 \end{bmatrix} \pm \begin{bmatrix} 1.5 \\ 0.27 \end{bmatrix}$$

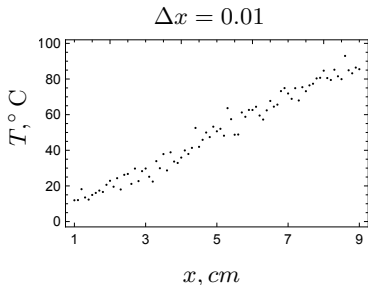
$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} -0.4 \\ 10.03 \end{bmatrix} \pm \begin{bmatrix} 1.0 \\ 0.18 \end{bmatrix}$$



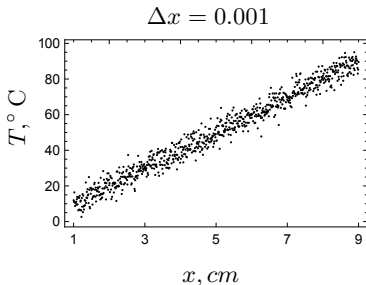


Mesh Refinement: Steps 3–4

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} -0.12 \pm 0.34 \\ 10.011 \pm 0.061 \end{bmatrix}$$



$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} 0.01 \pm 0.11 \\ 9.990 \pm 0.019 \end{bmatrix}$$

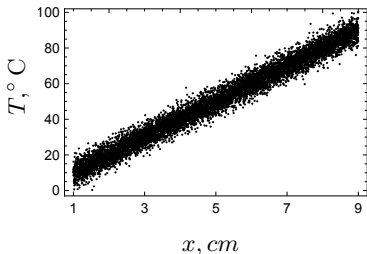




Mesh Refinement: Steps 5–6

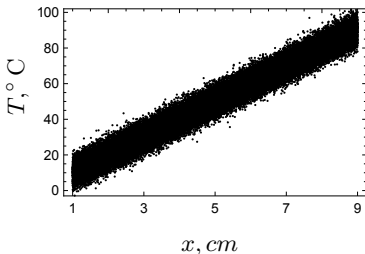
$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} -0.029 \pm 0.034 \\ 10.004 \pm 0.0061 \end{bmatrix}$$

$$\Delta x = 0.0001$$



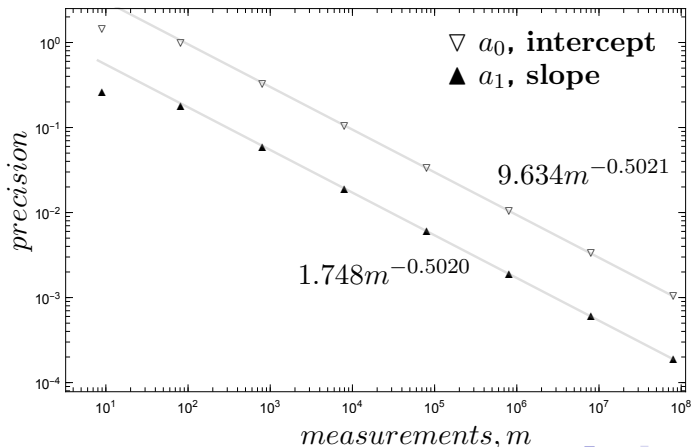
$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_p = \begin{bmatrix} -0.001 \pm 0.011 \\ 9.9994 \pm 0.0019 \end{bmatrix}$$

$$\Delta x = 0.00001$$



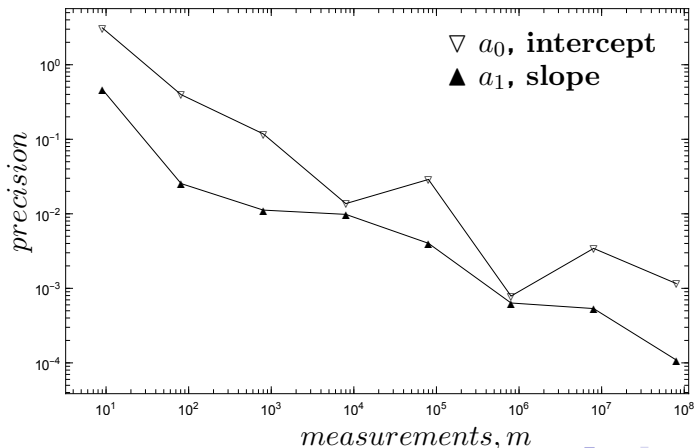


Mesh Refinement: Precision



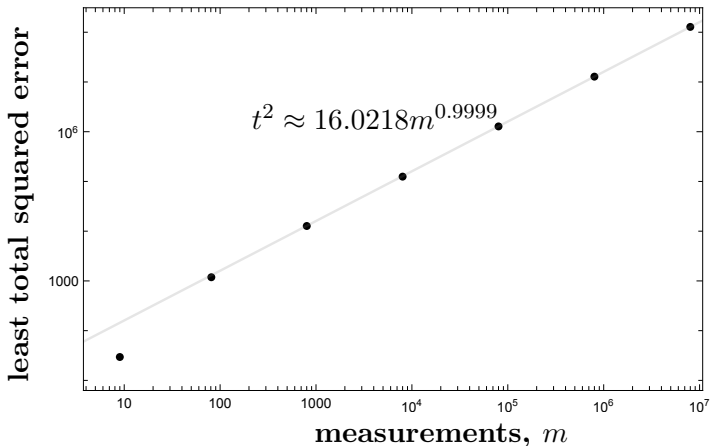


Mesh Refinement: Accuracy



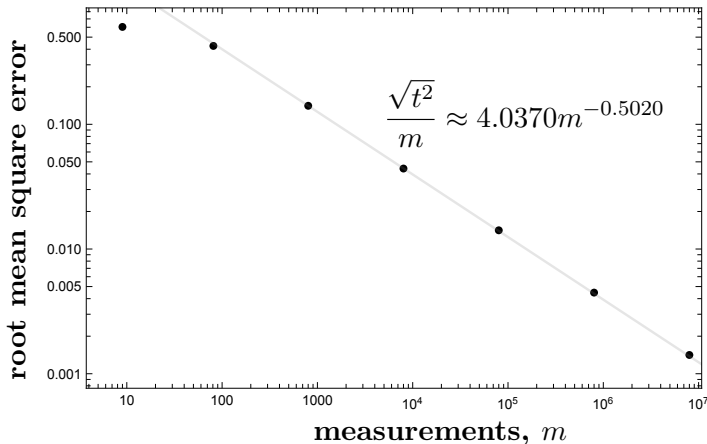


Mesh Refinement: Error





Mesh Refinement: Error





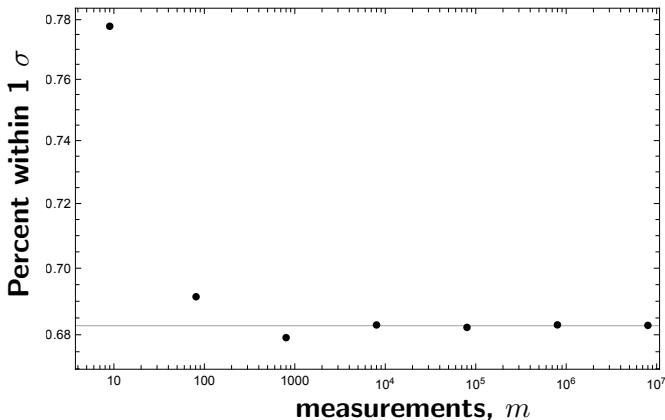
Area within 1σ

68–95–99.7 rule

$$\int_{-\sigma}^{\sigma} f(x; 0, \sigma) dx = 0.682689492$$

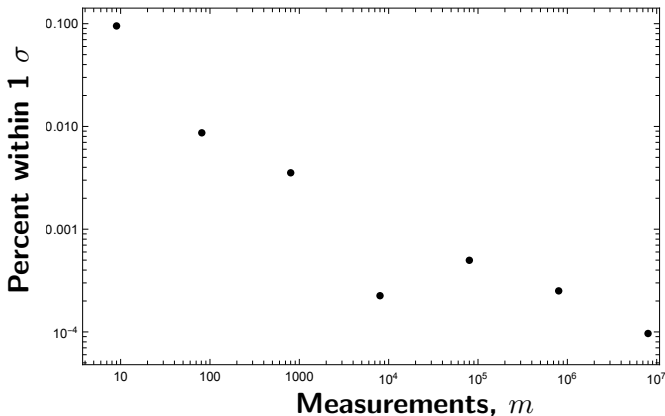


Mesh Refinement: Convergence Rate





Mesh Refinement: Measuring Standard Deviation





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Result
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Tinker

Seeing the Result
Interpolation
Convergence
McAninch Challenge
Perturbation Analysis



McAninch Challenge

Can we tell if this experiment was done in **Albuquerque**?



McAninch Challenge

- 1 Albuquerque average **elevation**: 1619 m
- 2 **Boiling point** of water at 1600 m: 94.4°C

How can we use this apparatus to determine if the water is boiling at 94.4°C ?



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McAninch Challenge

Without refining the apparatus, how do we **refine the solution**?



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McAninch Challenge

Let's insist that 100° and 94.4°C be **3 standard deviations** apart.



McAninch Challenge

Let's insist that 100° and 94.4°C be **3 standard deviations** apart.

Eqn. (38): $\sigma_{x=10} = (100 - 94.4) / 3 = 5.6 / 3 = 1.9$



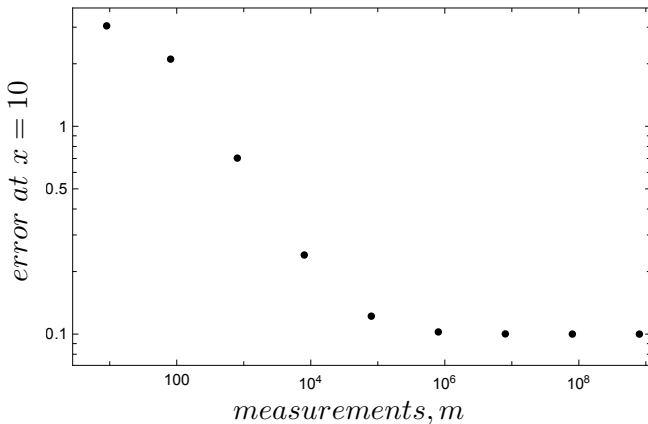
Solve for σ_0 and σ_1

Let's call the position error 0.01 cm.

$$\begin{aligned}\sigma_{x=10}^2 &= \sigma_0^2 + x^2 \sigma_1^2 + a_1^2 \sigma_x^2, \\ \sigma_{x=10}^2 &= \sigma_0^2 + 10^2 \sigma_1^2 + 10^2 (0.01)^2 = 1.9^2.\end{aligned}\tag{38}$$

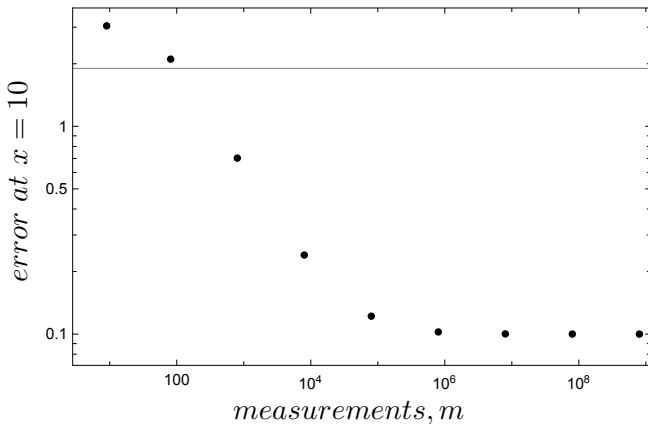


Device Characterization





Measurement Goal 1.9





Result

To use our code to **distinguish altitude**, we must make a **200 measurements**.



Perturbation Analysis Overview

We explore two forms of perturbation analysis:

- 1 Perturb Solution: Explore a δ -neighborhood of $M(a_p)$
- 2 Perturb Data: perturb a point in an ideal solution



Perturbation Analysis: Solution

How does the merit function change as the least squares solution is perturbed?

$$M(a_p + \delta) - M(a_p) \geq 0. \quad (39)$$



Perturbation Analysis: Solution

$$\begin{aligned}a_0 &\rightarrow a_0 + \delta, \\a_1 &\rightarrow a_1 + \delta.\end{aligned}\tag{40}$$



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Sotto Voce

Tedious algebra follows; **no magic** here.



Perturbation Analysis: Solution

$$\begin{aligned} M(a) &= \sum_{k=1}^m (T(x_k) - T_k)^2 = \sum_{k=1}^m (a_0 + a_1 x_k - T_k)^2 \\ &= (a_0 \mathbf{1} + a_1 x - T)^T (a_0 \mathbf{1} + a_1 x - T) \\ &= \langle a_0 \mathbf{1} + a_1 x - T, a_0 \mathbf{1} + a_1 x - T \rangle \end{aligned} \tag{41}$$



Perturbation Analysis: Solution

$$\begin{aligned}M(a) &= \langle r(a), r(a) \rangle = \langle a_0 \underset{1}{\mathbf{1}} + a_1 x - T, a_0 \underset{1}{\mathbf{1}} + a_1 x - T \rangle \\&= a_0^2 \underset{1}{\langle \mathbf{1}, \mathbf{1} \rangle} + a_1^2 \underset{2}{\langle x, x \rangle} + \underset{3}{\langle T, T \rangle} + 2a_0 a_1 \underset{4}{\langle \mathbf{1}, x \rangle} \\&\quad - 2a_0 \underset{5}{\langle \mathbf{1}, T \rangle} - 2a_1 \underset{6}{\langle x, T \rangle}\end{aligned}\tag{42}$$



Perturbation Analysis: Solution

	$M(a_p + \delta)$ perturbed	—	$M(a_p)$ unperturbed	=	ζ difference
1	$(a_0 + \delta)^2 \langle \mathbf{1}, \mathbf{1} \rangle$	—	$a_0^2 \langle \mathbf{1}, \mathbf{1} \rangle$	=	$(2a_0\delta + \delta^2) \langle \mathbf{1}, \mathbf{1} \rangle$
2	$(a_1 + \delta)^2 \langle x, x \rangle$	—	$a_1^2 \langle x, x \rangle$	=	$(2a_1\delta + \delta^2) \langle x, x \rangle$
3	$\langle T, T \rangle$	—	$\langle T, T \rangle$	=	0
4	$2(a_0 + \delta)(a_1 + \delta) \langle \mathbf{1}, x \rangle$	—	$2a_0a_1 \langle \mathbf{1}, x \rangle$	=	$2((a_0 + a_1)\delta + \delta^2) \langle \mathbf{1}, x \rangle$
5	$-2(a_0 + \delta) \langle T, T \rangle$	—	$-2a_0 \langle T, T \rangle$	=	$-2\delta \langle T, T \rangle$
6	$-2(a_1 + \delta) \langle x, T \rangle$	—	$-2a_1 \langle x, T \rangle$	=	$-2\delta \langle x, T \rangle$
Σ	$t^2 + \zeta$	—	t^2	=	ζ



Perturbation Analysis: Solution

$$\begin{aligned}\zeta &= \delta^2 (\langle x, x \rangle + 2\langle \mathbf{1}, x \rangle + \langle \mathbf{1}, \mathbf{1} \rangle) \\ &= \delta^2 \langle x + \mathbf{1}, x + \mathbf{1} \rangle \geq 0\end{aligned}\tag{43}$$



Perturbation Analysis: Solution Verification

- **Compute** a_0, a_1
- **Compute** t^2
- **Input** δ
- **Compute** $t^2 + \zeta$
- **Demonstrate** $\zeta = \delta^2 \langle x + 1, x + 1 \rangle$



Perturbation Analysis: Data

Next, take a data sequence with an **ideal solution** ($t^2 = 0$) and **perturb a single point**.



Perturbation Analysis: Data

Perturbation:

$$T_k \rightarrow T_k + \delta \quad (44)$$

Implies

$$\begin{aligned} \langle x, T \rangle &\rightarrow \langle x, T \rangle + x_k \delta, \\ \langle \mathbf{1}, T \rangle &\rightarrow \langle \mathbf{1}, T \rangle + \delta. \end{aligned} \quad (45)$$



Perturbation Analysis: Data

With modest effort equation 27 becomes

$$\begin{aligned}a_0 &\rightarrow a_0 + \frac{\delta}{\Delta} (\langle x, x \rangle - \langle \mathbf{1}, x \rangle x_k), \\a_1 &\rightarrow a_1 + \frac{\delta}{\Delta} (-\langle \mathbf{1}, x \rangle + \langle \mathbf{1}, \mathbf{1} \rangle x_k) .\end{aligned}\tag{46}$$



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The Best Fit May be Crap
Why Least Squares?
The Normal Equations
Notation



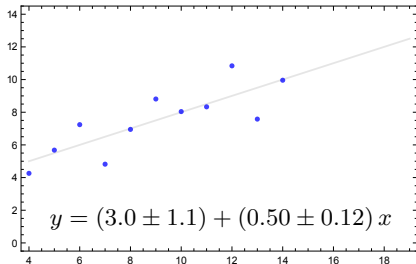
Nature Choses the Functional Form

The Best Fit May be Crap.

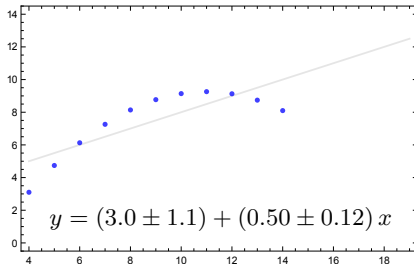


Anscombe's data set quartet I

Data set 1



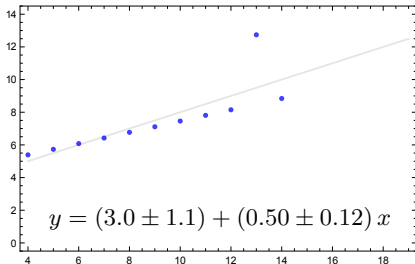
Data set 2



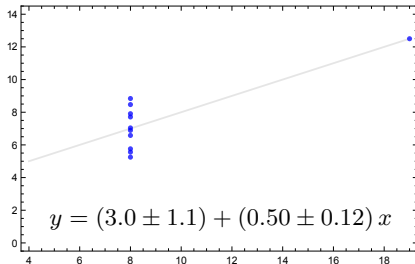


Anscombe's data set quartet II

Data set 3



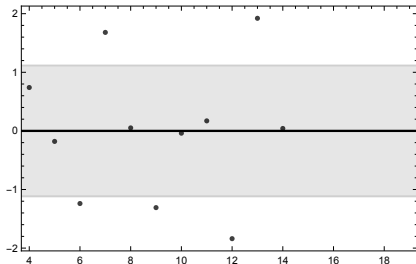
Data set 4



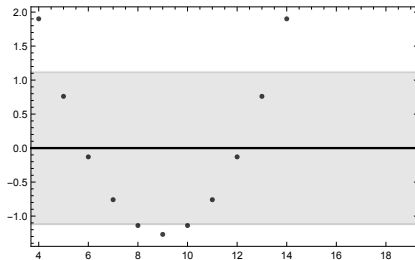


Anscombe's data set quartet I

Data set 1



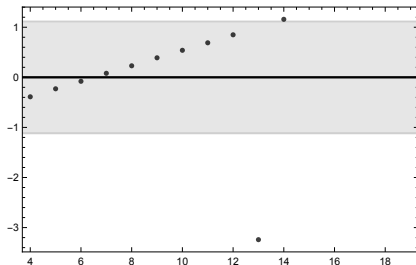
Data set 2



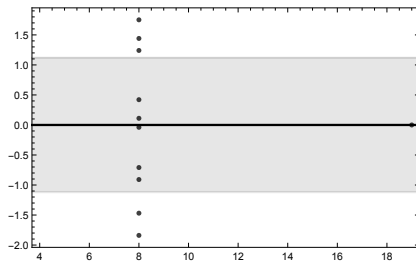


Anscombe's data set quartet II

Data set 3



Data set 4





Least Squares Does Not

- 1 Legitimize a bad model
- 2 Turn bad data into good results



Least Squares Does

- 1 Provide **quantitative** data for model **comparison**
- 2 Provide **best fit** for the chosen model



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Vector Norms

Norms quantify **length** and **proximity**.



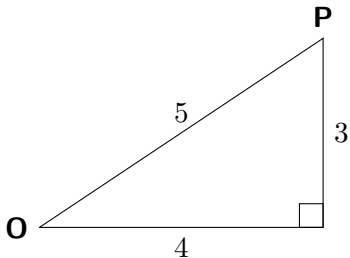
The Continuum of p norms

$$1 \leq p < \infty$$

$$\|x\|_p = \left(|x_1|^{1/p} + |x_2|^{1/p} + \cdots + |x_m|^{1/p} \right)^{1/p}$$



Norms as Path Length





Popular p norms

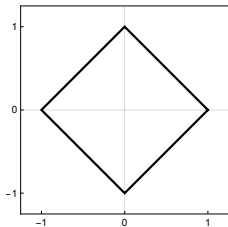
$$\begin{aligned}\|p\|_1 &= |x| + |y| &= 7 \\ \|p\|_2 &= \sqrt{x^2 + y^2} &= 5 \\ \|p\|_\infty &= \max(|x|, |y|) &= 4\end{aligned}\tag{47}$$



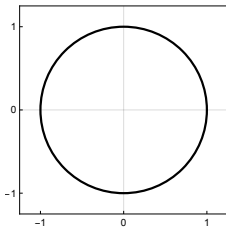
The Unit Circle in Different Norms

$$\|x + y\|_p = 1$$

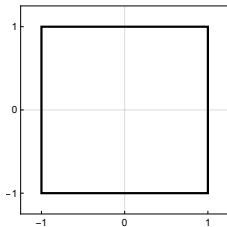
$p = 1$



$p = 2$



$p = \infty$





Why Use $\|\cdot\|_2$: Incorrect Answers

- 1 Herd instinct
- 2 Peer pressure
- 3 Precedent



Why Use $\|\cdot\|_2^2$: Correct Answers

- 1 Gauss–Markov Theorem
- 2 Differentiability



The Normal Equations in Numeric Computation





The Normal Equations in Numeric Computation

system

condition number

$$\mathbf{A} x = b \quad \kappa_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$$

$$\mathbf{A}^* \mathbf{A} x = \mathbf{A}^* b \quad \kappa_2(\mathbf{A}) = \left(\frac{\sigma_1}{\sigma_n} \right)^2$$



Problem
Model
Solution
Result
Discussion
Tinker

The Best Fit May be Crap
Why Least Squares?
The Normal Equations
Notation



The Normal Equations in Numeric Computation

Noisy problems become **unsolvable**



The Normal Equations in Numeric Computation

Linear system with 3-digit arithmetic¹

$$\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 1 & 2.01 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 9 \\ 3.01 \end{bmatrix} \quad (48)$$

$$\mathbf{x}_{LS} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (49)$$

¹Meyer, Example 4.5.1



The Normal Equations in Numeric Computation

Normal equations are **singular**²

$$\mathbf{A}^* \mathbf{A} = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \quad (50)$$

²e.g. row 2 = 2 × row 1



The Normal Equations in Theory

$$\mathbf{A}^* \mathbf{A} x = \mathbf{A}^* b$$

- ① Always consistent
- ② For $\rho = n$, solution is **unique**
- ③ For $\rho = n$, solution is $x_{LS} = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* b$



The Normal Equations in Theory

Proof: Equivalence of
 x_N , **normal equations solution** and
 x_{LS} , **least squares solution**



The Normal Equations in Theory

Proof: Outline

$$1 \quad t^2 = \| \mathbf{A}x_{LS} - b \|_2^2$$

definition



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|Ax_{LS} - b\|_2^2$

definition

2 $Ax_N - Ax_N$

propitious zero



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|\mathbf{A}x_{LS} - b\|_2^2$

definition

2 $\mathbf{A}x_N - \mathbf{A}x_N$

propitious zero

3 $t^2 = \|\mathbf{A}x_{LS} + \mathbf{A}x_N - \mathbf{A}x_N - b\|_2^2$

restatement of 1



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|\mathbf{A}x_{LS} - b\|_2^2$

definition

2 $\mathbf{A}x_N - \mathbf{A}x_N$

propitious zero

3 $t^2 = \|\mathbf{A}x_{LS} + \mathbf{A}x_N - \mathbf{A}x_N - b\|_2^2$

restatement of 1

4 $t^2 \leq \|\mathbf{A}x_N - b\|_2^2 + \|\mathbf{A}(x_{LS} - x_N)\|_2^2$

triangle inequality



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|\mathbf{A}x_{LS} - b\|_2^2$

2 $\mathbf{A}x_N - \mathbf{A}x_N$

3 $t^2 = \|\mathbf{A}x_{LS} + \mathbf{A}x_N - \mathbf{A}x_N - b\|_2^2$

4 $t^2 \leq \|Ax_N - b\|_2^2 + \|\mathbf{A}(x_{LS} - x_N)\|_2^2$

5 ...

definition

propitious zero

restatement of 1

triangle inequality

Miracle



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|Ax_{LS} - b\|_2^2$

2 $Ax_N - Ax_N$

3 $t^2 = \|Ax_{LS} + Ax_N - Ax_N - b\|_2^2$

4 $t^2 \leq \|Ax_N - b\|_2^2 + \|A(x_{LS} - x_N)\|_2^2$

5 ...

6 $t^2 = \|Ax_N - b\|_2^2 + \|A(x_{LS} - x_N)\|_2^2$

definition

propitious zero

restatement of 1

triangle inequality

Miracle

target (Pythagoras)



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|\mathbf{A}x_{LS} - b\|_2^2$

2 $\mathbf{A}x_N - \mathbf{A}x_N$

3 $t^2 = \|\mathbf{A}x_{LS} + \mathbf{A}x_N - \mathbf{A}x_N - b\|_2^2$

4 $t^2 \leq \|Ax_N - b\|_2^2 + \|\mathbf{A}(x_{LS} - x_N)\|_2^2$

5 ...

6 $t^2 = \|Ax_N - b\|_2^2 + \|\mathbf{A}(x_{LS} - x_N)\|_2^2$

7 $t^2 = t^2 + \|\mathbf{A}(x_{LS} - x_N)\|_2^2$

definition

propitious zero

restatement of 1

triangle inequality

Miracle

target (Pythagoras)

evaluation



The Normal Equations in Theory

Proof: Outline

1 $t^2 = \|Ax_{LS} - b\|_2^2$

2 $Ax_N - Ax_N$

3 $t^2 = \|Ax_{LS} + Ax_N - Ax_N - b\|_2^2$

4 $t^2 \leq \|Ax_N - b\|_2^2 + \|A(x_{LS} - x_N)\|_2^2$

5 ...

6 $t^2 = \|Ax_N - b\|_2^2 + \|A(x_{LS} - x_N)\|_2^2$

7 $t^2 = t^2 + \|A(x_{LS} - x_N)\|_2^2$

8 $\|A(x_{LS} - x_N)\|_2^2 = 0$

definition

propitious zero

restatement of 1

triangle inequality

Miracle

target (Pythagoras)

evaluation

conclusion



Notation Options

Examples of notation

- 1 Inner product form
- 2 Vector form
- 3 Scalar form



Exact Solution Notation: Inner Products

$$a_p = \Delta^{-1} \begin{bmatrix} \langle \mathbf{1}, y \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, y \rangle \\ \langle \mathbf{1}, \mathbf{1} \rangle \langle x, y \rangle - \langle x, \mathbf{1} \rangle \langle \mathbf{1}, y \rangle \end{bmatrix} \quad (27)$$

$$\Delta = \langle \mathbf{1}, \mathbf{1} \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, \mathbf{1} \rangle \quad (28)$$



Exact Solution Notation: Vectors

$$a_p = \Delta^{-1} \begin{bmatrix} (x^*x) (\mathbf{1}^T y) - (\mathbf{1}^T x) (x^*y) \\ (\mathbf{1}^T \mathbf{1}) (x^*y) - (x^* \mathbf{1}) (\mathbf{1}^T y) \end{bmatrix} \quad (27)$$

$$\Delta = (\mathbf{1}^T \mathbf{1}) (x^*x) - (\mathbf{1}^T x) (x^* \mathbf{1}) \quad (28)$$



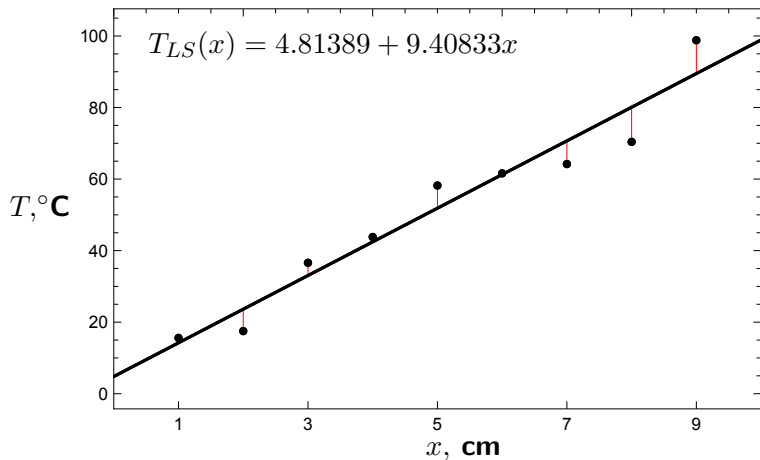
Exact Solution Notation: Scalar Form

$$a_p = \Delta^{-1} \left[\frac{\sum \bar{x}_k x_k \sum T_k - \sum x_k \sum \bar{x}_k T_k}{\sum 1 \sum \bar{x}_k T_k - \sum \bar{x}_k \sum T_k} \right] \quad (27)$$

$$\Delta = \sum 1 \sum \bar{x}_k x_k - \sum x_k \sum \bar{x}_k \quad (28)$$

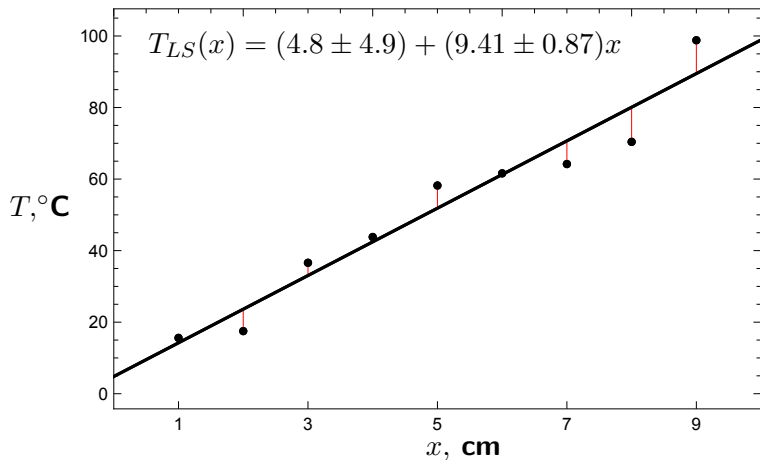


Data, Fit, Result, Residuals

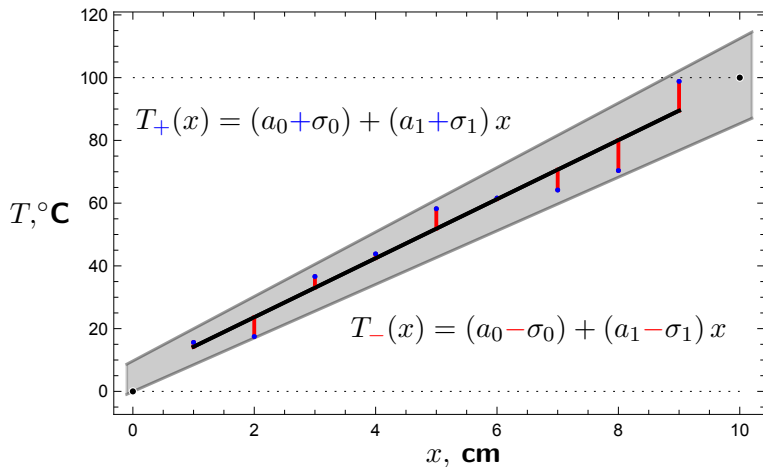




Data, Fit, Result, Residuals



Data, Fit, 1 σ Error



Linear Regression 101: Bevington Archetype

Daniel Topa
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Defense Threat Reduction Information Analysis Center (DTRIAC)

March 30, 2023